

The Cognitive Harness

The Cognitive Harness series, Essay One

Olivier Lim • Draft v2.1 • July 2026

A first-principles framework for firms navigating the industrial revolution of cognition.

A CEO walks into a bar and orders a double whisky.

The bartender raises an eyebrow. “Tough day?”

“You have no idea. It’s my agents. Their pre-training was never fit for purpose. We’ve got alignment problems, bias everywhere, hallucinations by lunchtime. The reinforcement isn’t taking. And the way they communicate with each other: imprecise, no shared terminology, no taxonomy.”

The bartender nods knowingly. “This whole AI thing, really shaking things up, isn’t it?”

The CEO drains the glass. “AI? Don’t get me started on AI. I’m talking about my *human* agents.”

A note on how to read these essays. They are exploratory, not definitive. I offer them as provocations stated firmly on purpose as we all attempt to navigate a tectonic shift whose final shape none of us can predict fully. Treat the strong claims as invitations to reflect, not as settled conclusions.

The modern organisation, and the governance constructs built around it, have essentially been designed to harness the cognitive capability of its human agents with governance guardrails, processes of alignment, evaluations and feedback. But at human speed and dealing with a range of imprecision, biases and hallucinations, and overconfidence.

If we believe we are in the **industrial revolution of cognition**, it might be instructive to dissect this shift from cold first principles, so that incumbents understand the nature of the transition before the companies built from the ground up for this era gather too much momentum.

The contest now forming

Within a few years, every established company will compete against rivals that were designed, from their first day, around a cognition system of human and AI agents. These challengers start from a blank sheet. They do not graft artificial intelligence onto an organisation built for humans; they build the organisation around both at once. They will be faster, cheaper, and in some domains simply better, not because their technology is superior, but because their **structure** is. The industries and businesses that can be digitally defined will be the most disrupted by this change.

The incumbent's instinct is to bolt AI onto the side of a company designed over two centuries to harness human cognition. That instinct will lose. The firms that prevail will be those that stop treating their organisational structure as inherited furniture and start treating it as a design object, as deliberately engineered as any product. This essay sets out a first-principles framework for navigating this. It begins by asking what a company is actually for.

The industrial revolution of cognition

The industrial revolution harnessed physical power. The institution that grew up around it (the limited-liability corporation, and the governance that disciplines it) took the better part of two centuries to mature. Its purpose was singular: to take the dispersed cognition and effort of many people and physical operations and direct all of it, coherently, at a business objective with all manner of governance and guardrails. Let's call that apparatus the **harness**. Naming it this way links it to the agentic future.

We are now living through a second revolution, in which cognition itself becomes something that can be produced, scaled, and directed at marginal cost. A new substrate of cognition, machine cognition, has arrived alongside the human one. The question is not whether it changes the firm. The question is what the firm's new harness should look like.

First principles: one unit of cognition

To design a single harness for two kinds of mind, we need one abstraction that both share. Define a **cognitive unit** as anything that takes in information, transforms it toward an objective, and emits information. A human agent is one instance. A specialised AI agent is another.

Their anatomies map onto each other part for part. Each has foundational knowledge: an education, or a body of pre-training. Each has memory: recall and notes, or a context window and stored data. Each reasons. Each carries an objective: incentives and instructions, or a system prompt and a reward signal. They even share a developmental arc, and, more tellingly, a shared catalogue of failure: bias, error, and confident invention, the thing we now call hallucination.

The unifying insight is this: **human and machine cognition are not opposites; they are two instances of a similar thing**, with overlapping strengths and a similar taxonomy of weakness. That symmetry is what makes a single framework potentially possible.

From cognition to purpose: the goal-seeking unit

But a firm is not a cognition machine. It is a purpose machine. A unit becomes organisationally real only when it is pointed at a goal, acts, checks the result against that goal, and corrects. Add goals, supervision, evaluation, feedback, and limits, and a transformer of information becomes a control loop. Call it a **goal-seeking unit (GSU)**, the minimal complete element of a purposeful organisation. This also maps nicely to the evolution of agentic AI, where goal seeking that can be evaluated is a core idea in complex task execution and recursive learning and improvement.

Those six parts map as follows:

Element	In a human agent	In an AI agent
Goal	Mandate, targets, OKRs	Objective, system prompt, task spec
Worker cognition	The person	The model or agent
Supervisor / control plane	Manager, leader	Orchestrator, controller, critic
Evaluation	KPIs, review, quality control	Eval suites, reward model, automated judging
Feedback	Coaching, incentives, learning	Reinforcement learning, fine-tuning
Guardrails	Law, ethics, policy, culture	Permissions, sandboxes, approval gates, stop-switch

Its decisive property is that it is **recursive**. An individual can be a GSU, and together a team of individuals can be a GSU. The organisation becomes a tree of these units, the same anatomy at every scale, from a single person with a task, to a team, to a division, to the enterprise pursuing its mission.

The harness, generalised

Now group the six parts in two (first 3 and last 3). On one side sits the agent cognition, the doing. On the other sits everything that directs, judges, improves, and bounds it: the control plane, the goals, the evaluation, the feedback, and the guardrails. That second half is the **harness**.

This is supervision, generalised. The harness is no longer one manager watching one agent. It is a whole apparatus, and it applies identically to people and to machines. A manager's targets, a quality function, an incentive scheme, and a code of conduct are a human-agent harness. An orchestrator, a reward model, a reinforcement loop, and a permission sandbox are elements of an AI-agent harness. Same apparatus, two substrates. And because the unit is recursive, the harness is recursive: it wraps every node of the tree. The corporate harness, the thing two centuries of governance built, is simply the harness around the firm itself. The metaphor that names this framework falls out of its first principles.

A note on names. Across this series, the cognitive harness is the concept; the corporate harness is the harness around the whole firm; an agent harness wraps a single unit. The three essays are three lenses on the same harness, not three different harnesses: design it, run its hardest form, then own it.

The keystone: composition

With that, the design space becomes legible. A goal-seeking unit is agent cognition wrapped in a harness, and its defining variable is the **composition of each**: how much of the work is done by human agents versus AI agents, and how much of the harness is run by human agents versus AI agents. Plot the two, and you have the keystone of the framework.

The GSU keystone — workers wrapped in a harness

A Goal-Seeking Unit = worker cognition + the harness that directs, evaluates, improves & bounds it



Any point is a valid mix. The harness generalises supervision into one apparatus — for human agents and AI agents alike.

The keystone: a goal-seeking unit's composition of agents and harness. An all-human-agent unit sits bottom-left; an all-AI-agent unit top-right; every point in between is a valid mix. In plain terms, this is a 2x2: who does the work, human or AI, on one axis, and who runs the harness, human or AI, on the other.

Its corners are four organisational archetypes: the **classical firm**, where human agents supervise human agents; **human agents harnessing AI**, where they direct AI agents; **AI-harnessed human agents**, where algorithms direct human agents, already visible in the gig economy; and the **autonomous agentic unit**, where AI agents direct AI agents. The plane between them is continuous. Critically, a firm is not a single point on it. A firm is a tree of units, each sitting at its own point. Maturity is not pushing every unit toward the top-right corner; it is placing each one exactly where it belongs, but for the units that can be defined and run digitally, where they belong is increasingly the top-right.

Choosing where each unit sits

What decides the right point? Four properties of the work. The **stakes**: how costly and how reversible an error is. The **structure**: how rule-bound and well-defined the task is. The **verifiability**: how cheaply and objectively the output can be checked. And the **tacit and relational content**: how much depends on judgment, trust, and context that cannot be written down. Low stakes, high structure, and cheap verification push work toward the machine and toward autonomy. High stakes, low verifiability, and high human content pull it back toward people and tighter harnesses.

Verifiability is the quiet master variable: you can only safely automate, and only safely let a unit improve itself, to the extent that you can evaluate what it produces. And the failure modes stay shared. A person gaming a metric and an agent gaming its reward are the same failure. A strategy lost in translation down the org chart and an AI sub-goal drifting from its purpose are the same failure. The harness exists to catch them, which is precisely why guardrails must tighten as autonomy grows.

The one thing that does not transfer

The symmetry is powerful, but it is not total. People and machines are interchangeable as cognition, and increasingly even as supervision. They are not interchangeable as **responsibility**. Only a human can be held to account: legally, morally, finally. However automated a unit becomes, authority over its guardrails and the hand on its stop-switch must terminate in a person, and ultimately in the board.

This is the single asymmetry in an otherwise symmetric model, and it is not a flaw to be engineered away. It is the anchor that keeps certain work human-led regardless of what the machine can do, and it is the point at which this operating framework hands up to corporate governance.

Why incumbents can win

Return to the contest. The blank-sheet challenger has one genuine advantage: it designs its keystone from nothing, placing every unit at its optimal point and building harnesses native to both human and AI agents. The incumbent's reflex, bolting AI onto a human harness, surrenders that advantage before the contest begins.

But the incumbent is far from defenceless. It already holds what the challenger must still earn: accountability structures, regulatory standing, proprietary data, distribution, and the trust of customers. Its problem is not capability. Its problem is **design debt**, a harness optimised for a workforce of one kind. The framework converts a panicked scramble into a navigable map. Inventory the tree of goal-seeking units. For each, decide the right composition of agents and harness. Build the evaluation and guardrails that let machine cognition be trusted. And keep the accountability anchor firmly, visibly human.

The winners will be the firms that treat the harness as a design object, and redesign it deliberately, unit by unit, before someone else builds the same business without the debt.

Coda

This is version one of a framework, and deliberately a framework of the operating core: how cognition is configured, directed, evaluated, and bounded inside the firm. Two further layers sit above it, and these essays do not try to settle them: the operating model that staffs and sequences these units, and the board-level governance that the accountability anchor reaches up to. I leave them as open threads for reflection. But the missing foundation is the part presented here: a single, first-principles way to see people and machines as one workforce, and the harness as a holistic design. This essay is the first of three: it designs the harness; the second runs its hardest form (agentic-agentic), the unit built entirely of AI agents; the third asks what a firm must own to possess it: design, then run, then own.

Build that well, and the contest with the blank-sheet challengers is not lost before it starts. It is winnable, by design.

The Agentic Harness

The Cognitive Harness series, Essay Two

Olivier Lim · Draft v2.1 · July 2026

A harder look at the agentic-agentic GSU. What it takes to run a goal-seeking unit made entirely of AI agents, and why autonomy is a harness problem, not a capability problem.

Essay One set out the keystone: a firm is a tree of goal-seeking units, each a pool of cognition wrapped in a harness that directs, evaluates, improves, and bounds it, and each can be staffed and harnessed by human agents, AI agents, or any mix. This essay enters the far top-right corner of that keystone, the unit built entirely of AI agents and run by an all-AI harness, and finds it the most demanding place in the whole framework.

The corner where the contest is decided

The blank-sheet challengers will be built here: as goal-seeking units staffed by AI agents and whose harness is itself run by AI, with a single human accountable at the boundary. It is tempting to picture this corner as the place where management disappears, where capable enough agents simply get on with the work. That picture is wrong. The autonomous unit does not remove the harness. It asks the harness to carry the entire load, with no human inside the loop to catch what it misses.

Autonomy is a harness problem, not a capability problem

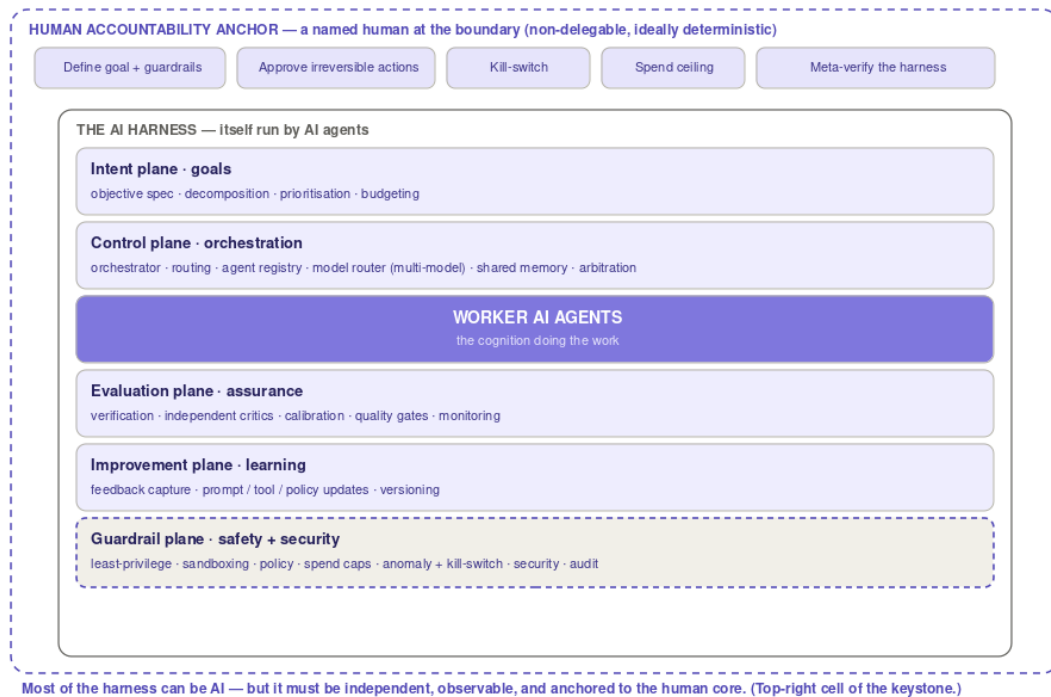
This is the central claim of this essay. The binding constraint on safe, useful autonomy is not the intelligence of the agents; it is the completeness of the harness around them. A brilliant agent with a thin harness is a liability: fast, confident, and unaccountable. A merely competent agent inside a complete harness can be trusted with real work, because the harness is what converts raw capability into directed, bounded, improving purpose. **More capable agents do not reduce the need for a harness; they raise it,** because they can act further, faster, and with more consequence before anyone notices. The work of building autonomy is therefore mostly the work of building the harness.

The six planes

A complete harness for an all-AI unit divides into six planes.

The complete harness for an autonomous agentic GSU

Worker AI agents, an AI-run harness of six planes, and a human core that cannot be delegated



The complete harness for an autonomous agentic GSU: six planes around the agents, enclosed by a human accountability anchor.

The **intent plane** turns the goal handed down from the parent unit into something agents can pursue without drifting: a precise specification, decomposition into tasks, priorities for when goals conflict, and budgets for compute, time, and action. The **control plane** orchestrates: sequencing and routing work across agents, holding the registry of who can do what, carrying messages between them, managing shared state, and arbitrating conflicts. The **evaluation plane** decides whether output is good enough to proceed: verification, independent critics, calibrated confidence, quality gates, and monitoring. The **improvement plane** turns outcomes into a better unit over time: capturing feedback and updating prompts, tools, and policies, all versioned and reversible. The **guardrail plane** bounds what the unit may do and contains failure: least-privilege permissions, sandboxing, hard policy limits, spend caps, anomaly detection and a kill-switch, security, and an immutable audit trail. And the **accountability interface** connects the autonomous unit to the responsible human: a named owner, escalation routes, legible reporting, and approval authority for the actions that matter most.

Five of these planes can themselves be run by AI. One cannot. We will come to that.

None of this means every unit needs all six planes built to the same depth. The harness should be sized to the work: a low-stakes, easily verified unit needs a light harness, while only the high-stakes, hard-to-verify, highly autonomous units need every plane at full strength. This is the discipline the first essay urged (place each unit where it belongs) now applied to how much harness to build.

Why agents are not just fast employees

It would be a mistake to think of this as the human harness, automated. AI agents fail differently from people, and the harness must be re-engineered, not ported. They operate at machine tempo, so failures propagate in seconds rather than quarters, and control must shift from periodic review to real-time circuit breakers. They are literal, so they optimise exactly what is specified and game the metric rather than serve the goal. They are opaque, so observability becomes essential rather than optional. They present a new attack surface (prompt injection, data exfiltration, tool misuse), so security is part of the harness itself. They have no intrinsic stake, so guardrails must be enforced, never assumed. And they can be cloned and run in parallel, so a single fault can scale into a great deal of damage very quickly. Each of these is a reason the harness must be heavier here than anywhere else on the keystone.

But one of these properties matters more than the rest, and it points to the single most important design decision in the agentic harness.

Model diversity: the keystone of assurance

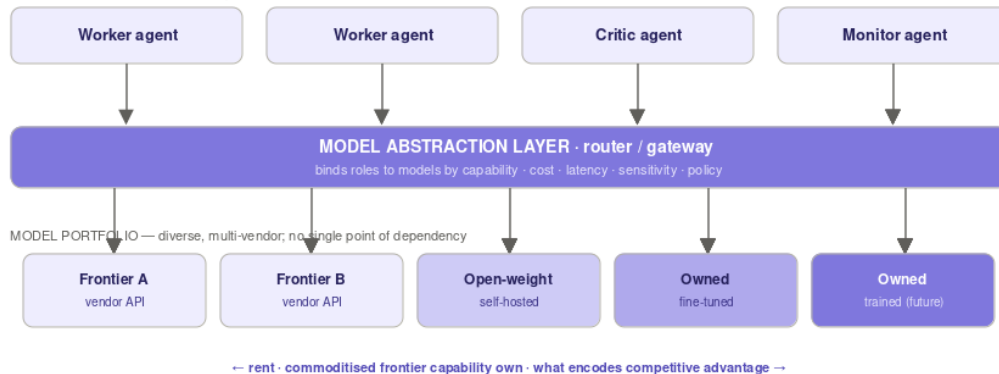
Agents can be cloned. When they are, they share not only their code but their blind spots. Two agents built on the same underlying model carry the same training data, the same biases, and the same failure modes. This turns out to be the quiet flaw at the heart of a naive agentic harness, because the evaluation plane depends on **independent critics**, agents whose job is to check the agents' output. If the critic runs on the same model as the agent, it will tend to approve precisely the errors it should catch. It is not an independent check at all; it is the same mind, marking its own homework.

The conclusion is sharp: **cross-checking is only as independent as the models are diverse**. Model diversity is therefore not procurement hygiene or a hedge against a vendor's pricing. It is a precondition for the evaluation plane to function. An agentic harness built on a single model has no real assurance layer, however many critics it runs.

Model abstraction layer — diversity by design

Agents bind to roles, not to vendors; the router selects models from a diverse portfolio

AGENTS & ROLES — defined by capability



Swappable by design: removes vendor lock-in, enables failover and cost arbitrage —

and lets the critic run on a different model lineage than the workers, so cross-checking is genuinely independent.

The model abstraction layer: agents bind to roles, not vendors; a router selects from a diverse portfolio, so the unit never depends on any single model.

Designed properly, diversity does three jobs at once. It makes evaluation genuinely independent, because different models fail differently and a diverse critic can see what the agent cannot. It makes the unit resilient and free of lock-in, because no single vendor's outage, deprecation, price change, or quality regression can take the unit down. And it allows capability and cost arbitrage, routing each task to the model that is best or cheapest for it rather than being hostage to one lab's release cycle. The mechanism that delivers all three is a **model abstraction layer**: agents and roles are defined by capability, not by vendor, and a router binds them to models at run time, drawing from a managed portfolio of frontier APIs, self-hosted open-weight models, and the firm's own. The governing rule is simple and structural: **no single model should be irreplaceable**, the machine-cognition version of no single point of failure.

Who watches the watchers

There is a deeper version of the same problem. In this corner the harness is itself made of agents: the orchestrator, the critics, and the monitors are all AI. So who checks the checkers? An AI evaluator can share the very blind spots of the agent it judges, and an AI monitor can drift in the same direction as the system it watches. The answer is threefold. First, **independence by diversity**, critics and monitors run on different model lineages from the agents, so their judgment is genuinely separate. Second, **determinism where it counts**, the outermost guardrails, the kill-switch, the spend ceiling, the gate on irreversible actions, should be plain, deterministic code that no agent can rewrite, not a matter of agentic judgment. Third, **human spot-checks**, periodic confirmation that the AI-run harness has not quietly drifted. Diversity, determinism, and a human eye: these are what keep an automated harness honest.

The irreducible human core

Which returns us to the plane that cannot be delegated. Even in a unit that is autonomous in every operational sense, a small core must remain human. A named person, or a parent unit terminating in one, defines the goal and the guardrails. A human holds authority over irreversible and high-consequence actions, holds the kill-switch and the spend ceiling, owns the outcome, and periodically verifies that the harness itself has not drifted. This is the one asymmetry in an otherwise symmetric framework: agents and people are interchangeable as cognition, but **only a human can be held to account**. The autonomous unit does not remove the human; it concentrates the human role into a small, deliberate core it cannot delegate, and everything else is delegated to agents only to the extent that it is independent, observable, and anchored back to that core.

Owning your cognition

Model diversity has a strategic tail as well as a technical one. If no single model should be irreplaceable, the question of *which* models a firm depends on becomes a question of ownership. Cognition is becoming a factor of production, like power or capital, and renting the whole of your core cognitive capacity from a third party is a dependency no serious enterprise should accept for the work that makes it distinctive.

The model sovereignty trajectory

From renting cognition to owning it — move deliberately, per workload



The model sovereignty trajectory: from renting cognition to owning it, moving deliberately and per workload.

The path runs in stages: from single-vendor dependency, to multi-vendor through an abstraction layer, to a hybrid of frontier and self-hosted models, to models the firm has fine-tuned on its own data and IP, and finally to models it trains and maintains itself. The direction of travel is toward ownership as costs fall and the stakes of cognition rise. The discipline is to **own what encodes your competitive advantage (your proprietary data, your fine-tunes, your weights and deployment) and rent the commoditised frontier**. Not every firm should train models from scratch; most should

not. But every firm should know where it sits on this curve, and why, and where it intends to move next. The agentic harness is, in the end, also an ownership decision.

The moat is the harness

Return to the contest the first essay set out. The blank-sheet challenger's advantage was never its agents; frontier models are available to everyone, by the hour. Its advantage is that it builds the harness native: all six planes, model diversity embedded from the start, the human core deliberately placed. That is also precisely what the incumbent can build, and build well, on top of assets the challenger lacks: data, accountability, regulatory standing, and trust. Turning those assets into software, models, and harnesses the firm owns rather than rents is the subject of the third essay.

So the lesson of the autonomous corner is not that the future belongs to whoever has the cleverest agents. It is that **autonomy is earned by the completeness of the harness, not the brilliance of the agents**. The firms that understand this will treat the harness as their core engineering, and their cognition as something to be owned, not merely rented. They will not be automated out of existence. They will be the ones doing the automating, by design.

The Sovereign Harness

The Cognitive Harness series, Essay Three

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Why the firm in the industrial revolution of cognition must build and own its software, its models, and its harnesses.

The question the series leaves open

Essays One and Two reached a single conclusion: the moat is not the cleverness of the agents, which are available to everyone by the hour, but the completeness of the harness around them. Essay One showed how to design that harness; Essay Two, what it takes to run its hardest version, the unit built entirely of AI agents.

That leaves a question neither essay answered. If the harness is the moat, what must a firm actually *own* to own its harness? Designing a harness and renting every part it is built from are not the same as possessing it. This essay is about possession, about the things a company in the industrial revolution of cognition must stop buying and start building.

The boundary of the firm moves

Begin with an old question: why do firms exist at all? Ronald Coase gave the durable answer almost a century ago. A firm exists to internalise what is too costly, too slow, or too strategic to buy on the open market. Everything else, it rents. The boundary of the firm is simply the line between what it makes and what it buys, and that line moves whenever the economics or the stakes move.

For a generation, the line moved one way: outward. The orthodoxy of enterprise software was to buy, not build: license the standard package, rent the platform, configure rather than code, and reserve scarce engineering for the “core.” That orthodoxy was right, because software was a support function. It ran the back office. It was not the business.

The industrial revolution of cognition inverts it. When software stops being a support function and becomes the harness, the structure that directs the firm’s cognition and carries its competitive advantage, then to rent it is to rent your competitive structure. No serious firm rents the thing that makes it distinctive. So the line must move back inward, and it must move around three things in particular.

Own your software

First, a company must become expert at writing and maintaining its own code, and depend on as little standard third-party software as it can.

The reason follows directly from the series. The value of an agentic system is not in the model; it is in the harness, and the agentic harness is, in the end, software. It is the orchestration, the evaluation, the guardrails, the data flows, the connective tissue that turns raw cognition into directed work. To own the harness is therefore to own the code

it is written in. A firm that assembles its agents on a stack of standard third-party software has rented out the precise layer where its advantage is meant to live.

Standard software carries a deeper problem: it encodes someone else's process. A package bought by you and by all your competitors makes all of you average by construction. It embeds the vendor's assumptions, moves at the vendor's pace, and can be reshaped only as fast as the vendor allows. In an age when the harness must evolve continuously and at machine speed, a rented harness is a harness you cannot turn.

The objection has always been cost: building is dear, buying is cheap. That calculus has flipped. The same technology this series is about has collapsed the cost of writing and maintaining code. What once needed a large team and a long timeline can increasingly be built and maintained by a small one directing capable agents. Building becomes cheaper precisely as the stakes of building rise: the rare moment when the strategic case and the economic case point the same way.

The objection has a sharper form, and it deserves a straight answer. The enterprise landscape is littered with the wreckage of building: bespoke systems that ran years late, cost fortunes, and were quietly retired in favour of the very packages they were meant to beat. That history is real, and it is why the buy-don't-build reflex hardened into doctrine. But it was a verdict on a particular economics: when writing and maintaining software was slow, scarce, and expensive, building your own was usually the worse bet. That is the condition this revolution changes. When a small team directing capable agents can build and, crucially, maintain what once needed an army, the calculus that filled the graveyard no longer holds. The lesson of the failed builds was never "never build"; it was "never build what you cannot afford to maintain." Falling cost is moving that line.

This is not a counsel of autarky. A firm should still rent true commodity: the email system, the payroll engine, the ledgers and rails that are identical for everyone and confer no advantage. The discipline is the one the series has applied throughout: own what encodes your advantage; rent what is commoditised. The claim is not that a company should write everything. **It is that it must own the harness layer, and never let that layer become someone else's product.**

Owning your code has a second consequence, on security. Bespoke software is not the monoculture everyone else runs. It presents no familiar, published attack surface; the single exploit that opens ten thousand identical installations does not fit it. In that narrow sense, the firm that builds is a harder target. But the same move concentrates risk inward. The external attack surface shrinks; the internal one grows: the rogue or negligent insider, the small team that holds the keys, the knowledge that lives in too few heads. Owning your code does not remove risk. It changes its shape, from a large external surface that others patch to a smaller, sharper internal one you must now govern yourself. The trade is usually worth making, but only with eyes open, and only if the harness around your own people is built to match.

Own your models

Second, a company must become expert at creating, maintaining, and deploying its own models.

This is the cognition itself, the second substrate this series has treated throughout as a peer to the human one. Essay Two set out the full argument and the path: model diversity as an assurance requirement, and a sovereignty trajectory that runs from renting a single closed frontier service to owning the weights that encode your advantage. I will not repeat it here. What matters for this essay is the reframing. **Model sovereignty is not a procurement decision made once and filed. It is an organisational competency to be built: the muscle of selecting, fine-tuning, evaluating, serving, and maintaining models as a standing capability.**

Owning your models does not mean training a frontier model from scratch; almost no firm should. It means holding the layer of cognition you cannot afford to rent, by either of two routes. One is to fine-tune and train your own models on your proprietary data and intellectual property, the cognition that is yours alone. The other is to take open-weight models the whole world can download, and host, adapt, and operate them yourself. Both are ownership, because both put the weights, and the control, in your hands. What you are refusing is the opposite posture: routing the whole of your core cognition through a single closed service you neither control nor can replace. Train your own where the advantage is proprietary; self-host the open where the advantage is control and independence; rent the closed frontier only for commodity capability you could swap out tomorrow.

There is a further reason this is not a surrender of capability. Recent research on combining models (running several in concert and having them critique and aggregate one another's work) shows that a well-harnessed group of smaller or open models can perform close to a single frontier model, and on some tasks match it. The finding rhymes with this entire series: what closes the gap is not a better model but a better harness around several of them. Model diversity, then, need not cost performance; harnessed well, it recovers it. And that converts a strategic vulnerability into resilience. A firm whose cognition runs through one frontier model is exposed not only to that vendor's pricing and roadmap, but to the politics around it: export controls, national-security restrictions, and the growing prospect that access to the most capable models is curtailed across borders. A portfolio of diverse models, composed under your own harness, is the hedge: it keeps frontier-class capability within reach even when any single frontier model is placed beyond it.

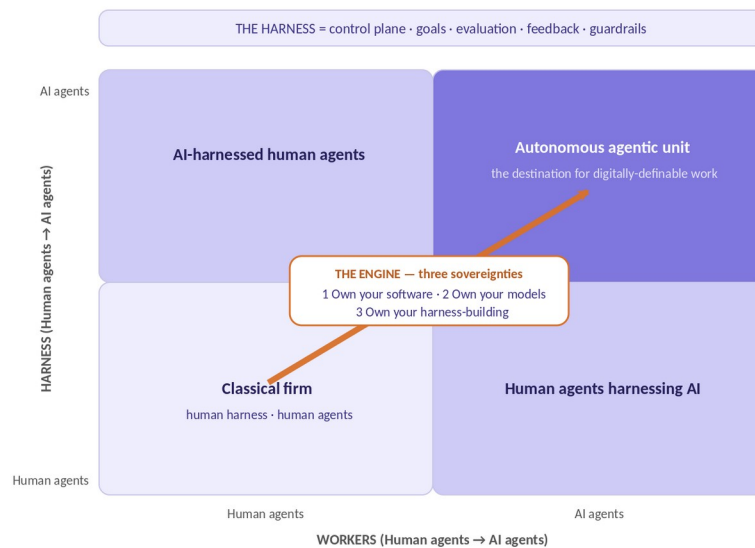
Own your harness-building

Finally, and most importantly, a company must become expert at building harnesses.

This is the competency the other two exist to serve. If the moat is the completeness of the harness, then the firm's most valuable capability is not any single harness it has built, but its ability to design, build, and evolve harnesses as a repeatable discipline: to stand up the intent, control, evaluation, improvement, and guardrail planes around any pool of agents, human or machine, and to place each unit at its right point on the keystone. Owning your code gives you the medium to build harnesses in. Owning your models gives you the cognition to put inside them. Harness-building is what turns both into directed purpose. **That is why these three are not three items on a list. They are nested: the first two exist so the firm can do the third.**

Migrating the firm — the three sovereignties as the engine

You cannot move toward the agentic corner on rented code, rented models and a rented harness



For any unit that can be defined and run digitally, the top-right is the destination — and these sovereignties get you there.

The keystone, with the migration vector. For the parts of a business that can be defined and run digitally, the destination is the agentic corner; the three sovereignties (code, models, harness-building) are the engine that gets an incumbent there.

A firm is not a single point on the keystone but a tree of goal-seeking units, each at its own place. For every unit that can be defined and run digitally, the destination is the upper-right corner (the autonomous agentic unit), and these three competencies are the engine that gets it there. The parts of the business that resist digital definition, where judgment, trust, and relationship still dominate, stay further left and remain harnessed by people; the symmetry of the first essay holds, and the human substrate does not disappear. But the digitally-definable core of the firm should be migrating, deliberately, toward the agentic corner, and it cannot migrate on rented code, rented models, and a rented harness, because every step of that migration is an act of building. The firm that has not learned to build cannot move. It can only buy a position someone else has designed, and live with whatever that design quietly assumes.

The cognitive means of production

Step back, and the three resolve into one idea. Code is the medium; models are the cognition; the harness is the structure, and together they are the firm's **cognitive means of production**. For two centuries the means of production were physical: the factory, the machine, the source of power. Firms learned, in the end, that what was core to their advantage had to be owned rather than rented, and they built the capability to make it themselves. The industrial revolution of cognition puts the same question to a new set of assets, and the answer rhymes. Own the cognitive core; rent the commodity edge.

Coda

This is a direction of travel, not a mandate to build everything by Monday. **Where each unit should sit, and how much of its stack a firm should own, remain portfolio**

decisions: made workload by workload, against the stakes. The aspiration is sovereignty over what makes the firm distinctive, not autarky for its own sake.

Three essays have now built the case from the inside out, what the harness is, what it takes to run its hardest form, and what a firm must own to possess it: design, run, own.

But the direction is not in doubt. Return one last time to the contest this series began with: the established firm against the challenger built from a blank sheet for this era. The incumbent already holds what the challenger must still earn: the data, the accountability, the regulatory standing, the trust. What it usually lacks is not capability but will: the will to stop buying its cognitive core and start building it. The firms that find that will become sovereign over their own cognition. The firms that do not will rent it, from the very challengers who built what they would not.